

PNEUMATICS: BUILDING AN AIR-POWERED ARM

Fun with Air and Movement



THE POWER OF YOUR BREATH

Did you know your breath
can move things? Let's find
out how!



WHAT IS PNEUMATICS?

Pneumatics means using air to make things move!

When you blow air into a balloon, it gets bigger. That same pushing power can move machines and tools.

Air is invisible but very strong!



HOW DOES AIR MAKE THINGS MOVE?

1

Air pushes
and pulls.

2

Air can fill
balloons or
tubes.

3

Moving air can
lift or push
objects.

PNEUMATICS VS HYDRAULICS



Uses air to make
things move

Uses liquid to
make things
move

Light, fast,
safe, and
clean!

Strong, slow,
and heavy



WHY USE PNEUMATICS?

Easy to control

Safe for kids

Works with
simple parts

Helps machines
move quickly

MEET THE MECHANICAL ARM

Our Project: The Air-Powered Arm
Let's build a robot arm that moves with air! Using simple materials and the power of your breath, we'll create something amazing together.



MATERIALS YOU NEED

Cardboard

Balloons

Straws

Air
Bladders

Ruler

Tape

Tubing

Scissors

Markers &
Labels

HOW TO BUILD YOUR ARM

Step 1: Cut two long strips of cardboard. These will be the "bones" of your arm.

Step 2: Use a pin or a straw to join the two cardboard strips so they can move like an elbow.

Step 3: Tape a balloon between the two strips of cardboard. Connect your plastic tubing to the mouth of the balloon and tape it tight!

Step 4: Attach the other end of the tube to a syringe or just prepare to use your own breath!

AIR POWER IN ACTION

When you blow air into the tube, the balloon inflates and pushes the arm parts. This makes the arm move up and down!



FUN PNEUMATICS FACTS QUIZ

1

What moves the
air-powered arm?
Answer: Air!

2

What do
hydraulics use?
Answer: Liquid!

3

Can you move things
with your breath?
Answer: Yes!

YOUR TURN TO TRY!

Use air to make your own moving arm. Have fun and explore!

